

Fixed Points I: Fixed Points over Multi-Bundle Manifolds

A conditional functional-analytic framework with spectral-bundle projections

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Part I of VI in the Fixed Point series. As both the cornerstone of the Modal Triplet Theory (MTT) collection and a stand-alone development, the series is intended to function simultaneously as a basis and as a self-contained study. Each paper in the series builds upon its predecessors, extending the fixed-point framework step by step.

Abstract

We develop a conditional fixed-point framework for gradient flows on internal fiber products of bundles over a base manifold Y of dimension $d := \dim Y$. Given smooth bundles $\pi_n : B_n \rightarrow Y$ with compact fibers F_n of uniformly bounded geometry, we construct spectral-bundle projections onto the joint harmonic sector and study gradient flows generated by coercive analytic energies $C : H^1(B_{\text{int}}) \rightarrow \mathbb{R}$ on the internal product

$$B_{\text{int}} := B_1 \times_Y \cdots \times_Y B_k.$$

We establish from first principles: (i) precise geometry and anisotropic Sobolev structure; (ii) uniform H^1 bounds for the coherent projector via Riesz-spectral calculus under bounded geometry; (iii) analytic-semigroup well-posedness and smoothing for Φ_t ; (iv) existence of projected fixed points by a variational method on the coherent sector; alternatively by Schauder (compact base with base-regularized smoothing) or by Darbo using the Kuratowski measure under a confining base potential; (v) uniqueness either from strong convexity on the coherent sector or via an inertial-manifold reduction under a clean spectral gap condition; (vi) quantitative Galerkin approximation for the uniformly elliptic steady problem with Céa-type quasi-optimal error; (vii) examples (with explicit constants) and Łojasiewicz-Simon convergence. All results are analytic/control statements under the displayed hypotheses. The Riemannian base is not automatically physical Lorentzian spacetime, the flow parameter is not automatically physical time, and the abstract fiber product does not by itself select the canonical MTT internal topology.

Revision note for this edition

Supersedes. *Fixed Points I: Fixed Points over Multi-Bundle Manifolds*, version 5.

Reason. The earlier proof chain needed explicit control geometry, anisotropic function spaces, projector bounds, compactness mechanisms, and separate hypotheses for existence, uniqueness, rate, and approximation.

Resolution. Version 6 repairs the spectral-calculus and semigroup estimates, states valid Schauder and Darbo routes, corrects the confining-space norm, and gives a Céa-type Galerkin theorem for the elliptic steady problem.

Retained result. Conditional projected fixed-point existence, coherent-sector uniqueness routes, and convergence remain valid under the displayed analytic hypotheses.

Remaining boundary. The abstract Riemannian control geometry, flow parameter, and fiber product do not select physical spacetime or the q79 internal carrier.

Contents

Revision note for this edition	1
Standing Series Axiom (shared across the FP series)	2
1 Geometry of the internal product and standing hypotheses	3
1.1 Internal fiber product	3
1.2 Bounded geometry and uniform spectral gap	4
1.3 Function spaces and anisotropic norms	4
2 Projectors via fiberwise spectral calculus	5
3 Gradient flow: well-posedness and smoothing	5
3.1 Energy, decomposition, and basic assumptions	5
3.2 Projected map and identification of equilibria	6
3.3 Detailed semigroup bound underpinning $M_1(t)$	7
4 Existence of equilibria and projected fixed points	7
4.1 Variational existence on the coherent sector	7
4.2 Alternative route: Schauder (compact base) and base-regularization	7
4.3 Darbo condensing on noncompact bases	8
5 Uniqueness of equilibria	8
5.1 Route (A): strong convexity / strong monotonicity on the coherent sector	8
5.2 Route (B): inertial manifold reduction	9
6 Galerkin approximation for the steady state	9
7 Examples and counterexamples	10
7.1 Compact fibers with fixed geometry: circles and tori have $\lambda_1 > 0$	10
7.2 How the uniform gap λ^* can fail	11
7.3 Effect of base potentials	11
7.4 Worked example with explicit constants: $Y = S^1$, $F(y) = S^1_{R(y)}$	11
8 Auxiliary estimates for spectral-bundle projectors	11
9 Łojasiewicz–Simon convergence of the gradient flow	11
10 Singular perturbation and the limit $\varepsilon \downarrow 0$	12
11 Related work and novelty	12

Standing Series Axiom (shared across the FP series)

Assumption 0.1 (Series Axiom: bounded geometry + uniform spectral gap + coherent projection).

- (SA1) **Base geometry.** (Y, g_Y) is a smooth, complete Riemannian manifold. When uniform constants involving base derivatives are required, we additionally assume Y has bounded geometry (uniform bounds on curvature and injectivity radius on the region of interest).
- (SA2) **Fiber bounded geometry.** Each bundle $\pi_n : B_n \rightarrow Y$ has compact fibers $F_n(y)$ whose Riemannian geometry is uniformly bounded in y (Assumption 1.3 below).
- (SA3) **Uniform fiber spectral gap.** The nonnegative fiber Laplacians $\Delta_{F_n(y)}$ have a uniform first positive eigenvalue bound $\lambda^* > 0$ (Assumption 1.4).
- (SA4) **Coherent projector.** The fiberwise Riesz projector Π_{coh} onto the joint harmonic sector is well-defined and bounded on the Sobolev scales used (Section 2).
- (SA5) **Coherence invariance (when invoked).** For equilibrium identification and coherent-sector variational arguments we assume $N(\text{Ran}\Pi_{\text{coh}}) \subset \text{Ran}\Pi_{\text{coh}}$ (Assumption 3.6).

1 Geometry of the internal product and standing hypotheses

1.1 Internal fiber product

Let Y be a smooth d -dimensional manifold with complete Riemannian metric g_Y . Let $\pi_n : B_n \rightarrow Y$ be smooth bundles with compact fibers $F_n(y)$.

Remark 1.1 (Control geometry versus physical spacetime). The base (Y, g_Y) is the Riemannian geometry used by the elliptic and parabolic estimates in this paper. Identifying it with a Cauchy slice, Euclideanized background, or control geometry associated with a Lorentzian spacetime requires a separate realization theorem. No Lorentzian causal or dynamical conclusion is inferred here.

The *internal bundle product* (fiber product over Y) is

$$B_{\text{int}} := B_1 \times_Y \cdots \times_Y B_k := \{(b_1, \dots, b_k) : \pi_1(b_1) = \cdots = \pi_k(b_k) \in Y\},$$

with projection $\pi : B_{\text{int}} \rightarrow Y$, $\pi(b_1, \dots, b_k) = y$.

Universal property. B_{int} comes with projections $\text{pr}_n : B_{\text{int}} \rightarrow B_n$ such that $\pi_n \circ \text{pr}_n = \pi_m \circ \text{pr}_m$ for all n, m . For any manifold M with maps $f_n : M \rightarrow B_n$ satisfying $\pi_1 \circ f_1 = \cdots = \pi_k \circ f_k$, there is a unique $f : M \rightarrow B_{\text{int}}$ with $\text{pr}_n \circ f = f_n$.

We equip each fiber

$$F(y) := \pi^{-1}(y) \cong F_1(y) \times \cdots \times F_k(y)$$

with a smooth product-type Riemannian metric $g_{F(y)}$ varying smoothly with y under bounded-geometry assumptions below, and equip B_{int} with the induced product-type metric.

Remark 1.2 (Scope of the fiber-product model). The displayed product is an abstract analytic construction. In an MTT realization with overlapping vertical structures, or nested structures after explicit inclusion maps have been supplied—in particular when there is a shared central circle—one must work on the selected six-dimensional internal space with the corresponding joint vertical operators; the shared structure must not be counted as an additional independent factor. The theorems below apply after that model supplies the required bounded geometry, joint projector, and gap hypotheses.

1.2 Bounded geometry and uniform spectral gap

Assumption 1.3 (Bounded geometry of fibers). There exist $K_0, \iota_0 > 0$ and an integer $m \geq 2$ such that for all $y \in Y$ and each $n = 1, \dots, k$, the fiber $(F_n(y), g_{F_n(y)})$ has $|\text{sec}| \leq K_0$, $\text{inrad} \geq \iota_0$, and the fiber metric depends C^m -smoothly on y with all y -derivatives up to order m bounded uniformly on Y (in bounded-geometry coordinate charts on the base, when needed).

Nonnegative Laplacians. Let $\Delta_{F_n(y)}$ denote the *nonnegative* Laplace–Beltrami operator on $F_n(y)$ (so $\Delta \geq 0$ and $\ker \Delta$ are the fiberwise harmonic modes). Define the fiber operator on the product fiber $F(y)$ by

$$\Delta_F(y) := \sum_{n=1}^k \Delta_{F_n(y)}.$$

Assumption 1.4 (Uniform fiber spectral gap). There exists $\lambda^* > 0$ such that

$$\lambda^* \leq \inf_{y \in Y} \min_{1 \leq n \leq k} \lambda_1(\Delta_{F_n(y)}),$$

where $\lambda_1(\Delta_{F_n(y)})$ is the first *positive* eigenvalue of the nonnegative fiber Laplacian. Moreover, $\dim \ker \Delta_{F_n(y)}$ is constant in y for each n (hence the harmonic bundle ranks are constant).

1.3 Function spaces and anisotropic norms

Let $L^2 := L^2(B_{\text{int}})$ with the product volume measure. Let ∇_F denote the gradient along fibers. Define the anisotropic (fiber) Sobolev space

$$H_F^1 := \{\Psi \in L^2 : \nabla_F \Psi \in L^2\}, \quad \|\Psi\|_{H_F^1}^2 := \|\Psi\|_{L^2}^2 + \|\nabla_F \Psi\|_{L^2}^2.$$

We use $H_0 := L^2$ and $H_F := H_F^1$ as the primary Banach spaces for mapping properties.

Definition 1.5 (Full Sobolev space). We write

$$H^1(B_{\text{int}}) := \{\Psi \in L^2(B_{\text{int}}) : \nabla_Y \Psi \in L^2, \nabla_F \Psi \in L^2\},$$

with norm

$$\|\Psi\|_{H^1}^2 := \|\Psi\|_{L^2}^2 + \|\nabla_Y \Psi\|_{L^2}^2 + \|\nabla_F \Psi\|_{L^2}^2.$$

Remark 1.6 (Why H_F^1 is the correct space for fiber coercivity). The fiber operator Δ_F controls $\|\nabla_F \Psi\|_{L^2}$ but does not control $\|\nabla_Y \Psi\|_{L^2}$. Thus a Poincaré-type inequality driven solely by Δ_F fails on the full $H^1(B_{\text{int}})$, while it holds naturally in H_F^1 . This is the key coercivity correction used throughout the series.

Coherent sector notation. Let Π_{coh} be the coherent projector (defined in Section 2) and set

$$H_{\text{coh}} := \text{Ran}(\Pi_{\text{coh}}) \cap H^1(B_{\text{int}}), \quad Q := \text{Id} - \Pi_{\text{coh}}.$$

2 Projectors via fiberwise spectral calculus

For each $y \in Y$ define the self-adjoint nonnegative operator

$$A(y) := \Delta_F(y) \quad \text{on } L^2(F(y)), \quad \text{dom}(A(y)) = H^2(F(y)).$$

By Assumption 1.4, $\sigma(A(y)) \cap (0, \lambda^*) = \emptyset$ uniformly in y .

Let Γ be the positively oriented circle of radius $\lambda^*/2$ centered at 0. Define the Riesz projector onto $\ker A(y)$ by

$$\Pi(y) := \frac{1}{2\pi i} \int_{\Gamma} (z - A(y))^{-1} dz.$$

Define Π_{coh} on $L^2(B_{\text{int}})$ fiberwise by

$$(\Pi_{\text{coh}}\Psi)(\cdot, y) := \Pi(y)\Psi(\cdot, y).$$

Lemma 2.1 (Uniform boundedness of Π_{coh} on L^2 and H_F^1). *Under Assumptions 1.3–1.4, there exists $C_{\Pi} \geq 1$ such that*

$$\|\Pi_{\text{coh}}\|_{L^2 \rightarrow L^2} \leq 1, \quad \|\Pi_{\text{coh}}\|_{H_F^1 \rightarrow H_F^1} \leq C_{\Pi}.$$

In the y -independent flat case one may take $C_{\Pi} = 1$.

Remark 2.2 (On $\text{Ran}(\Pi_{\text{coh}})$ for scalar fields). If the fibers are connected and Π_{coh} projects scalar functions onto $\ker \Delta_F$, then $\text{Ran}(\Pi_{\text{coh}})$ consists of fiberwise constants and $\nabla_F(\Pi_{\text{coh}}\Psi) = 0$, so $\|\Pi_{\text{coh}}\Psi\|_{H_F^1} = \|\Pi_{\text{coh}}\Psi\|_{L^2} \leq \|\Psi\|_{L^2}$ (hence $C_{\Pi} = 1$). The general $C_{\Pi} \geq 1$ form is kept for vector-valued settings (e.g. bundles of forms/spinors) where $\ker \Delta$ may have rank > 1 .

Lemma 2.3 (Boundedness on full H^1). *Under Assumptions 1.3–1.4 (and base bounded-geometry control when y -derivatives are invoked), Π_{coh} extends to a bounded map $H^1 \rightarrow H^1$, hence $\text{Ran}(\Pi_{\text{coh}})$ is closed in H^1 .*

Remark 2.4 (Proof references for spectral projectors). Uniform control of the Riesz–spectral projector on Sobolev scales follows from the Riesz–Dunford formula together with parameter-dependent elliptic regularity on bounded-geometry families; see, for example, [4, 2].

Remark 2.5 (Automatic commutation on the domain). Since $\Pi(y)$ projects onto $\ker A(y)$, we have $A(y)\Pi(y) = 0$ and $\Pi(y)A(y) = 0$ on $\text{dom}(A(y))$. Thus $[A, \Pi_{\text{coh}}] = 0$ on $\text{dom}(A)$, and by density/continuity on L^2 . Since Π_{coh} is a Riesz projector defined via the functional calculus of A , it commutes with A and hence with e^{-tA} .

3 Gradient flow: well-posedness and smoothing

3.1 Energy, decomposition, and basic assumptions

Let $C : H^1(B_{\text{int}}) \rightarrow \mathbb{R}$ be a real-analytic functional such that its L^2 -gradient decomposes as

$$\nabla C(\Psi) = A\Psi + N(\Psi),$$

where $A = \Delta_F$ acts fiberwise (nonnegative, self-adjoint) and

$$N : H^1(B_{\text{int}}) \rightarrow L^2(B_{\text{int}})$$

is locally Lipschitz on bounded sets (with Lipschitz constant L on the relevant sublevel set).

The gradient flow is

$$\partial_t \Psi(t) = -A\Psi(t) - N(\Psi(t)), \quad \Psi(0) = \Psi_0 \in L^2(B_{\text{int}}). \quad (3.1)$$

Remark 3.1 (Stabilization parameter). Here t parametrizes the analytic gradient/stabilization flow. It is not identified with physical Lorentzian time. A physical evolution operator $U(t_2, t_1)$ and its relation, if any, to this flow must be defined separately.

Assumption 3.2 (Well-posedness and smoothing). For every $\Psi_0 \in L^2$, (3.1) has a unique global mild solution $\Psi \in C([0, \infty); L^2) \cap C((0, \infty); H^1)$. For each $t > 0$, the time- t map $\Phi_t : \Psi_0 \mapsto \Psi(t)$ maps L^2 into H_F^1 and is locally Lipschitz as a map $L^2 \rightarrow H_F^1$; in particular, for some $M_1(t)$,

$$\|\Phi_t(\Psi_1) - \Phi_t(\Psi_2)\|_{H_F^1} \leq M_1(t)e^{Lt}\|\Psi_1 - \Psi_2\|_{L^2}.$$

Proposition 3.3 (Energy dissipation and forward invariance of sublevels). *Along sufficiently regular solutions of the L^2 -gradient flow (3.1), the energy is nonincreasing and*

$$C[\Psi(t)] + \int_0^t \|\nabla C(\Psi(s))\|_{L^2}^2 ds = C[\Psi_0] \quad (t \geq 0).$$

In particular, every energy sublevel $\{\Psi : C[\Psi] \leq M\}$ is forward invariant. For semilinear analytic-semigroup flows this identity can be justified by standard approximation arguments; see, e.g., [1, 3].

$$D_M := \{\Psi \in H^1(B_{\text{int}}) : C[\Psi] \leq M\}.$$

Assumption 3.4 (Forward invariance of energy sublevels). For each $M > 0$, the energy sublevel $D_M = \{\Psi \in H^1(B_{\text{int}}) : C[\Psi] \leq M\}$ is forward invariant under the flow Φ_t used in the Darbo argument (in particular, under the base-regularized flow if $\varepsilon > 0$ is employed).

Remark 3.5 (Semigroup/ODE splitting). Decompose $L^2 = \text{Ran}(Q) \oplus \text{Ran}(\Pi_{\text{coh}})$. On $\text{Ran}(Q)$ the sectorial operator $A|_{\text{Ran}(Q)} \geq \lambda^*$ generates an analytic semigroup with the standard $t^{-1/2}e^{-\lambda^*t}$ bound for $A^{1/2}e^{-tA}$. On $\text{Ran}(\Pi_{\text{coh}})$ the flow reduces to an ODE $\partial_t u = -\Pi_{\text{coh}}N(u)$ (when coherence invariance holds).

Assumption 3.6 (Coherence invariance). We say coherence invariance holds if

$$N(\text{Ran}(\Pi_{\text{coh}})) \subset \text{Ran}(\Pi_{\text{coh}}).$$

Under this condition the coherent sector is invariant under the flow.

3.2 Projected map and identification of equilibria

Define

$$T_t := \Pi_{\text{coh}} \circ \Phi_t : L^2 \rightarrow \text{Ran}(\Pi_{\text{coh}}).$$

Proposition 3.7 (Projected fixed points are equilibria). *Assume coherence invariance (Assumption 3.6). If $\Psi^* \in \text{Ran}(\Pi_{\text{coh}})$ satisfies $T_t(\Psi^*) = \Psi^*$ for some $t > 0$, then $\Phi_s(\Psi^*) = \Psi^*$ for all $s \geq 0$, and $\nabla C(\Psi^*) = 0$.*

Proof. Coherence invariance implies $\Phi_s(\Psi^*) \in \text{Ran}(\Pi_{\text{coh}})$ for all $s \geq 0$, hence $T_t(\Psi^*) = \Phi_t(\Psi^*) = \Psi^*$. Applying the energy identity of Proposition 3.3 on $[0, t]$ gives

$$0 = C[\Phi_t(\Psi^*)] - C[\Psi^*] = - \int_0^t \|\nabla C(\Phi_s(\Psi^*))\|_{L^2}^2 ds.$$

The integrand therefore vanishes, so the orbit is stationary and $\nabla C(\Psi^*) = 0$. $\square$

Remark 3.8 (Without coherence invariance). If $N(\text{Ran}\Pi_{\text{coh}}) \not\subset \text{Ran}\Pi_{\text{coh}}$, then a fixed point of $T_t = \Pi_{\text{coh}}\Phi_t$ need not be a true equilibrium of Φ_t . In that case T_t can still be used to build invariant sets, but not to identify equilibria.

3.3 Detailed semigroup bound underpinning $M_1(t)$

Lemma 3.9 (Semigroup bound underpinning $M_1(t)$). *Let $Q := \text{Id} - \Pi_{\text{coh}}$. Then for all $t > 0$,*

$$\|A^{1/2}e^{-tA}Q\|_{L^2 \rightarrow L^2} \leq C t^{-1/2} e^{-\lambda^* t}, \quad \|e^{-tA}\Pi_{\text{coh}}\|_{L^2 \rightarrow L^2} = 1.$$

Consequently, for all $\Psi_1, \Psi_2 \in L^2$,

$$\|\Phi_t(\Psi_1) - \Phi_t(\Psi_2)\|_{H_F^1} \leq C(1 + t^{-1/2} e^{-\lambda^* t}) e^{Lt} \|\Psi_1 - \Psi_2\|_{L^2}.$$

In particular, one may take

$$M_1(t) := C(1 + t^{-1/2} e^{-\lambda^* t}).$$

Lemma 3.10 (Lipschitz continuity of T_t). *For each $t > 0$ there exists $K(t)$ such that for all $\Psi_{1,2} \in L^2$,*

$$\|T_t(\Psi_1) - T_t(\Psi_2)\|_{H_F^1} \leq C_{\Pi} K(t) e^{Lt} \|\Psi_1 - \Psi_2\|_{L^2}.$$

Moreover, compactness of T_t requires base regularization (see Section 4.2) or another mechanism that provides compactness in the base variable.

4 Existence of equilibria and projected fixed points

4.1 Variational existence on the coherent sector

Assumption 4.1 (Coercivity and weak lower semicontinuity). *$C : H^1(B_{\text{int}}) \rightarrow \mathbb{R}$ is bounded below, coercive on $H^1(B_{\text{int}})$, and sequentially weakly lower semicontinuous.*

Theorem 4.2 (Variational existence). *Assume coherence invariance (Assumption 3.6) and Assumption 4.1. Then there exists $\Psi^* \in H_{\text{coh}}$ such that $\nabla C(\Psi^*) = 0$. In particular, $T_t(\Psi^*) = \Psi^*$ for all $t \geq 0$.*

4.2 Alternative route: Schauder (compact base) and base-regularization

To obtain compactness of the time- t map in the *base* variable, introduce a base-regularized generator

$$A_{\varepsilon} := \Delta_F + \varepsilon \Delta_Y, \quad \varepsilon \in (0, 1],$$

and consider the corresponding flow

$$\partial_t \Psi = -A_{\varepsilon} \Psi - N(\Psi).$$

Assumption 4.3 (Base-regularized well-posedness). *For some $\varepsilon \in (0, 1]$ and $\delta > 0$, the flow with generator A_{ε} is globally well-posed and, for every $t > 0$, maps bounded subsets of L^2 into bounded subsets of $H^{1+\delta}(B_{\text{int}})$. This is a standing regularity hypothesis to be verified for each nonlinearity and model.*

Lemma 4.4 (Compactness for compact bases). *If Y is compact and Assumption 4.3 holds, then for each $t > 0$, $\Phi_t : L^2 \rightarrow H^1$ is compact on bounded sets. Consequently $T_t = \Pi_{\text{coh}} \Phi_t$ is compact on bounded sets in H^1 .*

Proof. The total fibered space is compact when Y and the fibers are compact. The Rellich embedding $H^{1+\delta} \hookrightarrow H^1$ is then compact. Compose this embedding with the smoothing in Assumption 4.3 and the bounded map $\Pi_{\text{coh}} : H^1 \rightarrow H^1$ from Lemma 2.3. $\square$

Theorem 4.5 (Schauder fixed point). *Assume Y is compact and Assumption 4.3 holds. Let $D \subset H_{\text{coh}}$ be nonempty, closed, bounded, and convex, and assume $T_t(D) \subset D$ for some $t > 0$. Then T_t has a fixed point in D .*

4.3 Darbo condensing on noncompact bases

Definition 4.6 (Kuratowski measure of noncompactness). For a bounded set B in a metric space, $\alpha(B)$ is the infimum of $r > 0$ such that B can be covered by finitely many sets of diameter $< r$.

Assumption 4.7 (Confinement). There exists a smooth $V : Y \rightarrow [0, \infty)$ with $V(y) \rightarrow \infty$ as $r(y) := \text{dist}_Y(y, y_0) \rightarrow \infty$ for some basepoint $y_0 \in Y$, and a constant $K \geq 0$ such that

$$C[\Psi] \geq \int_Y V(y) \|\Psi(\cdot, y)\|_{L^2(F(y))}^2 dy - K.$$

Lemma 4.8 (Uniform tail smallness). *Under Assumption 4.7, for each $M > 0$ and $\epsilon > 0$ there exists $R > 0$ such that*

$$\int_{\{r(y) > R\}} \|\Psi(\cdot, y)\|_{L^2(F(y))}^2 dy \leq \epsilon^2 \quad \text{whenever } \Psi \in H^1(B_{\text{int}}) \text{ and } C[\Psi] \leq M.$$

Lemma 4.9 (Compact-local / small-tail decomposition). *Assume Assumptions 4.1, 4.3, and 4.7, and fix $t > 0$. Let $\chi_R \in C_c^\infty(Y)$ satisfy $0 \leq \chi_R \leq 1$, $\chi_R \equiv 1$ on $\{r \leq R\}$ and $\chi_R \equiv 0$ on $\{r \geq R + 1\}$, and write $\pi : B_{\text{int}} \rightarrow Y$ for the base projection.*

Define the output cutoff maps

$$K_R(\Psi_0) := (\chi_R \circ \pi) \Phi_t(\Psi_0), \quad S_R(\Psi_0) := (1 - \chi_R \circ \pi) \Phi_t(\Psi_0).$$

Then $\Phi_t = K_R + S_R$. Moreover:

- (i) *For each $M > 0$, the map $K_R : D_M \rightarrow L^2(B_{\text{int}})$ is compact.*
- (ii) *If D_M is forward invariant under Φ_t , then for each $M > 0$,*

$$\sup_{\Psi_0 \in D_M} \|S_R(\Psi_0)\|_{L^2} \rightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Remark 4.10 (Why this fixes the “tail linearity” issue). Tail smallness follows either from Proposition 3.2 for the original gradient flow, or from Assumption 3.3 when working with the base-regularized flow used in the Darbo argument.

Proposition 4.11 (Condensing property on energy sublevels). *Assume Assumptions 4.1, 4.3, and 4.7, and fix $t > 0$ and an energy level $M > 0$. On the invariant sublevel $D_M := \{\Psi : C[\Psi] \leq M\}$, the map Φ_t is Kuratowski-condensing in the sense of Sadovskii: for every bounded set $B \subset D_M$ with $\alpha(B) > 0$ one has $\alpha(\Phi_t(B)) < \alpha(B)$. Consequently, $T_t = \Pi_{\text{coh}} \circ \Phi_t$ is condensing in L^2 on any T_t -invariant subset of $D_M \cap \text{Ran}(\Pi_{\text{coh}})$, because Π_{coh} is an L^2 -orthogonal projection and hence has norm one.*

Theorem 4.12 (Darbo–Sadovskii fixed point). *Assume Assumptions 3.6, 4.1, 4.3, and 4.7. Fix $t > 0$ and an energy level $M > 0$. Let $D \subset D_M \cap \text{Ran}(\Pi_{\text{coh}})$ be nonempty, L^2 -closed, bounded, convex, and invariant under T_t . Then $T_t : D \rightarrow D$ has a fixed point. By Proposition 3.7, that fixed point is an equilibrium of the gradient flow.*

5 Uniqueness of equilibria

5.1 Route (A): strong convexity / strong monotonicity on the coherent sector

Assumption 5.1 (Strong monotonicity on H_{coh}). There exists $\mu > 0$ such that for all $u, v \in H_{\text{coh}}$,

$$\langle \nabla C(u) - \nabla C(v), u - v \rangle_{L^2} \geq \mu \|u - v\|_{L^2}^2.$$

Theorem 5.2 (Uniqueness on the coherent sector). *Under Assumptions 3.6 and 5.1, $C|_{H_{\text{coh}}}$ has a unique minimizer Ψ^* , which is the unique equilibrium in H_{coh} . Moreover,*

$$\|\Phi_t(u) - \Phi_t(v)\|_{L^2} \leq e^{-\mu t} \|u - v\|_{L^2} \quad \forall u, v \in H_{\text{coh}}.$$

5.2 Route (B): inertial manifold reduction

Let P_0 be a finite-rank spectral-bundle projection and $Q_0 := \text{Id} - P_0$.

Assumption 5.3 (Spectral gap and Lipschitz nonlinearity). Assume there exists $\gamma > 0$ such that

$$\text{dist}(\sigma(A|_{\text{Ran}(Q_0)}), \sigma(A|_{\text{Ran}(P_0)})) \geq \gamma, \quad \|N(u) - N(v)\|_{L^2} \leq L\|u - v\|_{L^2}.$$

Theorem 5.4 (Inertial manifold). *If $L < \gamma/4$, there exists a finite-dimensional C^1 inertial manifold $M = \{\xi + G(\xi) : \xi \in \text{Ran}(P_0)\}$ that is invariant and exponentially attracting.*

Remark 5.5 (References). Inertial manifold constructions for semilinear parabolic equations under a spectral-gap condition are classical; see, for example, [6, 3].

Assumption 5.6 (Reduced strong monotonicity). There exists $\mu > 0$ such that the reduced vector field on $\text{Ran}(P_0)$ is strongly monotone with constant μ .

Theorem 5.7 (Uniqueness via inertial manifold). *Under Assumptions 5.3 and 5.6, the flow has a unique equilibrium $\Psi^* \in M$, exponentially attracting on M . If additionally coherence invariance holds and $\text{Ran}(\Pi_{\text{coh}}) \subset \text{Ran}(P_0)$, then $\Psi^* \in H_{\text{coh}}$ and agrees with the coherent-sector variational minimizer.*

Example 5.8 (IM works while convexity fails). Let $F = S_R^1$ and $N(u) = u^3 - u$ (Allen–Cahn). N is not convex so Assumption 5.1 fails. On $\{\|u\|_{H^1} \leq \rho\}$ one has $\|N'(u)\| \leq 3\rho^2 + 1 =: L(\rho)$, and γ is comparable to the spectral gap $\lambda^* = R^{-2}$. Choosing ρ so that $L(\rho) < \gamma/4$ yields the inertial manifold and uniqueness on it.

6 Galerkin approximation for the steady state

We approximate equilibria by discretizing a uniformly elliptic steady problem. To ensure coercivity (and uniqueness) we assume the elliptic part includes a strictly positive zeroth-order term (mass or confinement).

Assumption 6.1 (Coercive elliptic steady operator). Let $\varepsilon > 0$ and assume the steady operator has the form

$$\tilde{A}_\varepsilon := \Delta_F + \varepsilon \Delta_Y + W,$$

where $W \geq w_0 > 0$ is a measurable multiplication operator and the associated closed quadratic form has domain

$$V_\varepsilon := \{w \in H^1(B_{\text{int}}) : W^{1/2}w \in L^2\}.$$

Examples include $W \equiv m_0^2$ on compact Y and a confining potential $W = V(y) \rightarrow \infty$ on a noncompact base.

We consider the steady equation

$$\tilde{A}_\varepsilon \Psi + N(\Psi) = 0. \tag{6.1}$$

Definition 6.2 (Energy norm for $\tilde{A}_\varepsilon$). Under Assumption 6.1, define for $w \in V_\varepsilon$

$$\|w\|_\varepsilon^2 := \|\nabla_F w\|_{L^2}^2 + \varepsilon \|\nabla_Y w\|_{L^2}^2 + \int_{B_{\text{int}}} W|w|^2.$$

Lemma 6.3 (Strong monotonicity in the energy norm). Assume Assumption 6.1. Suppose N is L -Lipschitz in L^2 on the set of interest and $L < w_0$. Let $F := \tilde{A}_\varepsilon + N : V_\varepsilon \rightarrow V_\varepsilon^*$. Then for all u, v in that set,

$$\langle F(u) - F(v), u - v \rangle_{V_\varepsilon^*, V_\varepsilon} \geq \|u - v\|_\varepsilon^2 - L \|u - v\|_{L^2}^2 \geq \mu_\varepsilon \|u - v\|_\varepsilon^2,$$

where $\mu_\varepsilon := 1 - L/w_0 > 0$.

Theorem 6.4 (Céa-type quasi-optimal error in the energy norm). Assume Lemma 6.3. Let $V_N \subset V_\varepsilon$ be a finite-dimensional Galerkin space and let Ψ^* solve $\tilde{A}_\varepsilon \Psi + N(\Psi) = 0$. Let $\Psi_N \in V_N$ satisfy the Galerkin condition

$$\langle \tilde{A}_\varepsilon \Psi_N + N(\Psi_N), v \rangle_{V_\varepsilon^*, V_\varepsilon} = 0 \quad \forall v \in V_N.$$

Then Ψ_N is unique (for each N) and

$$\|\Psi^* - \Psi_N\|_\varepsilon \leq \frac{1 + L/w_0}{1 - L/w_0} \inf_{v \in V_N} \|\Psi^* - v\|_\varepsilon.$$

If W is also bounded above, the norm $\|\cdot\|_\varepsilon$ is equivalent to H^1 , hence this yields an H^1 error bound with an additional explicit norm-equivalence constant. For a confining unbounded W , the result is instead an estimate in the weighted form space V_ε .

Proof. The quadratic-form part has continuity constant one in $\|\cdot\|_\varepsilon$. Moreover,

$$|\langle N(u) - N(v), w \rangle| \leq L \|u - v\|_{L^2} \|w\|_{L^2} \leq \frac{L}{w_0} \|u - v\|_\varepsilon \|w\|_\varepsilon.$$

Thus F has continuity constant $1 + L/w_0$ and strong-monotonicity constant $1 - L/w_0$. Galerkin orthogonality and the standard nonlinear Céa argument give the displayed ratio. Strong monotonicity also gives uniqueness of the finite-dimensional solution. $\square$

Remark 6.5. In Theorem 6.4 the Lipschitz constant L is taken with respect to the L^2 norm. The conversion to the energy norm $\|\cdot\|_\varepsilon$ uses the inequality

$$\|u\|_{L^2} \leq w_0^{-1/2} \|u\|_\varepsilon,$$

which explains the appearance of the factor $(1 + L/w_0)$ in the Céa constant.

7 Examples and counterexamples

7.1 Compact fibers with fixed geometry: circles and tori have $\lambda_1 > 0$

Proposition 7.1. If F is compact, then the first positive Laplace eigenvalue $\lambda_1(\Delta_F) > 0$.

Corollary 7.2 (Circle/torus examples). On S_R^1 , $\sigma(\Delta) = \{(k/R)^2 : k \in \mathbb{Z}\}$, so $\lambda_1 = R^{-2} > 0$. Similarly for flat tori T^m one has $\lambda_1 > 0$.

7.2 How the uniform gap λ^* can fail

If radii degenerate (e.g. $R(y) \rightarrow \infty$) or fibers become noncompact, then λ^* can drop to 0, destroying uniform estimates.

7.3 Effect of base potentials

A smooth confining potential $V : Y \rightarrow [0, \infty)$ added to C improves tail coercivity (Assumption 4.7) but does not change the fiber spectra.

7.4 Worked example with explicit constants: $Y = S^1$, $F(y) = S_{R(y)}^1$

Assume $R \in C^1(S^1)$ with $0 < R_{\min} \leq R(y) \leq R_{\max}$ and $|R'(y)| \leq \rho$.

(a) **First eigenvalue:** $\lambda^* = \inf_y \lambda_1(\Delta_{F(y)}) = 1/R_{\max}^2$.

(b) **H^1 projector bound (one explicit model):**

with $A(y) = \Delta_{F(y)} = -R(y)^{-2} \partial_\theta^2$ one computes $\partial_y A(y) = 2R'(y)R(y)^{-3} \partial_\theta^2$ and hence $\|\partial_y A(y)\|_{H^2 \rightarrow L^2} \leq 2\rho R_{\min}^{-3}$. Using the Riesz formula and $\|(z - A)^{-1}\| \leq 2/\lambda^*$ for $|z| = \lambda^*/2$ yields a concrete

$$\|\Pi\|_{H^1 \rightarrow H^1} \leq 1 + \sup_y \|\partial_y \Pi(y)\|_{L^2 \rightarrow L^2} \leq 1 + \frac{4\rho R_{\max}^2}{R_{\min}^3}.$$

(c) **IM threshold:** taking $P_0 = \Pi$ gives $\gamma = \lambda^* = R_{\max}^{-2}$, and the IM criterion is $L < \gamma/4 = 1/(4R_{\max}^2)$.

8 Auxiliary estimates for spectral-bundle projectors

Theorem 8.1 (Explicit H_F^1 bound for Π_{coh}). *Under Assumptions 1.3–1.4 there exists*

$$C_\Pi \leq 1 + \frac{C(K_0, \iota_0, m)}{\lambda^*}$$

such that $\|\Pi_{\text{coh}}\|_{H_F^1 \rightarrow H_F^1} \leq C_\Pi$. If the fibers are y -independent and flat, $C_\Pi = 1$.

Lemma 8.2 (Norm equivalence on the harmonic bundle). *On $\text{Ran}(\Pi_{\text{coh}})$ one has $\nabla_F \Psi = 0$ in the scalar connected-fiber case, hence $\|\Psi\|_{H_F^1} = \|\Psi\|_{L^2}$. More generally, on $\text{Ran}(\Pi_{\text{coh}})$ all Sobolev norms are uniformly equivalent by elliptic regularity on the harmonic bundle.*

9 Łojasiewicz–Simon convergence of the gradient flow

Assumption 9.1 (Analyticity and Palais–Smale). C is real-analytic and satisfies a Palais–Smale condition on energy sublevels: if $C[\Psi_n]$ is bounded and $\|\nabla C(\Psi_n)\|_{L^2} \rightarrow 0$, then (Ψ_n) has a convergent subsequence in H^1 . We also assume the orbit under consideration remains in a bounded energy sublevel on which the gradient flow is precompact in H^1 and the local Łojasiewicz–Simon inequality is valid near every cluster point.

Theorem 9.2 (Łojasiewicz–Simon convergence). *Assume Assumption 9.1. For any solution $\Psi(t)$ of (3.1) with initial data in an energy sublevel, there exists Ψ_∞ with $\nabla C(\Psi_\infty) = 0$ and*

$$\lim_{t \rightarrow \infty} \|\Psi(t) - \Psi_\infty\|_{H^1} = 0.$$

If coherence invariance holds and $\Psi(0) \in \text{Ran}(\Pi_{\text{coh}})$, then $\Psi_\infty \in \text{Ran}(\Pi_{\text{coh}})$.

Remark 9.3 (Reference). The underlying Łojasiewicz–Simon inequality and convergence mechanism originate with [5].

Corollary 9.4 (Exponential rate under nondegeneracy). *If, in addition, the limiting equilibrium is nondegenerate in the relevant phase space—equivalently, the local Łojasiewicz–Simon exponent is $1/2$, for example because the linearized Hessian has a positive spectral gap after quotienting exact symmetries—then $\|\Psi(t) - \Psi_\infty\|_{H^1} \leq Ce^{-\eta t}$ for some $\eta > 0$. Uniqueness by itself does not imply an exponential rate.*

10 Singular perturbation and the limit $\varepsilon \downarrow 0$

Let Ψ_ε solve the base-regularized steady equation

$$(\Delta_F + \varepsilon \Delta_Y) \Psi_\varepsilon + N(\Psi_\varepsilon) = 0.$$

Assumption 10.1 (Uniform bounds). Assume coherence invariance and $\sup_{\varepsilon \in (0,1]} \|\Psi_\varepsilon\|_{H^1} < \infty$. Assume either Y is compact or a confining potential provides compactness on energy sublevels. Assume also that N is sequentially demicontinuous along this compactness class: whenever $u_j \rightharpoonup u$ in H^1 and $u_j \rightarrow u$ in L^2 , then $N(u_j) \rightharpoonup N(u)$ in H^{-1} . This holds for the standard subcritical Nemytskii nonlinearities but must be checked in each model.

Theorem 10.2 (Convergence as $\varepsilon \downarrow 0$). *Under Assumption 10.1 and, in addition, either*

- (i) *the Q -sector Lipschitz constant satisfies $L < \lambda^*$, or*
- (ii) *N is weakly monotone on the Q -sector: $\langle QN(u) - QN(v), Q(u - v) \rangle_{L^2} \geq 0$ for all u, v ,*

there exists a subsequence $\varepsilon_j \downarrow 0$ with $\Psi_{\varepsilon_j} \rightarrow \Psi^$ in L^2 and weakly in H^1 , where*

$$\Delta_F \Psi^* + N(\Psi^*) = 0, \quad \text{and necessarily } \Psi^* \in \text{Ran}(\Pi_{\text{coh}}).$$

Proof. The uniform H^1 bound and the compactness hypothesis give a subsequence with the stated weak- H^1 /strong- L^2 convergence. The term $\varepsilon_j \Delta_Y \Psi_{\varepsilon_j}$ tends to zero in H^{-1} , while the fiber form passes to the limit weakly and demicontinuity identifies the nonlinear limit. This yields the limiting steady equation.

Let $p = \Pi_{\text{coh}} \Psi^*$ and $q = Q\Psi^*$. Coherence invariance gives $QN(p) = 0$. Testing the Q equation against q gives

$$\lambda^* \|q\|_{L^2}^2 \leq -\langle Q(N(p + q) - N(p)), q \rangle.$$

Under alternative (i), the right side has absolute value at most $L\|q\|_{L^2}^2$, so $L < \lambda^*$ forces $q = 0$. Under alternative (ii), the nonlinear term is nonnegative on the left-hand side and again $q = 0$. Thus $\Psi^* \in \text{Ran}(\Pi_{\text{coh}})$. $\square$

11 Related work and novelty

Our existence and regularity statements are classical in spirit, but three aspects appear new or at least not treated in this precise combination: (i) spectral-bundle projections with uniform constants on fibered products with bounded geometry (Theorem 8.1), including explicit constants in a worked model; (ii) a condensing proof on noncompact bases via a compact-local/small-tail decomposition (Lemma 4.9) tied directly to confinement; (iii) a clean link between projected fixed points and true equilibria (Proposition 3.7) via coherence invariance.

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